

[DC] Towards a User-Centric Framework for Evaluating Locomotion Interfaces

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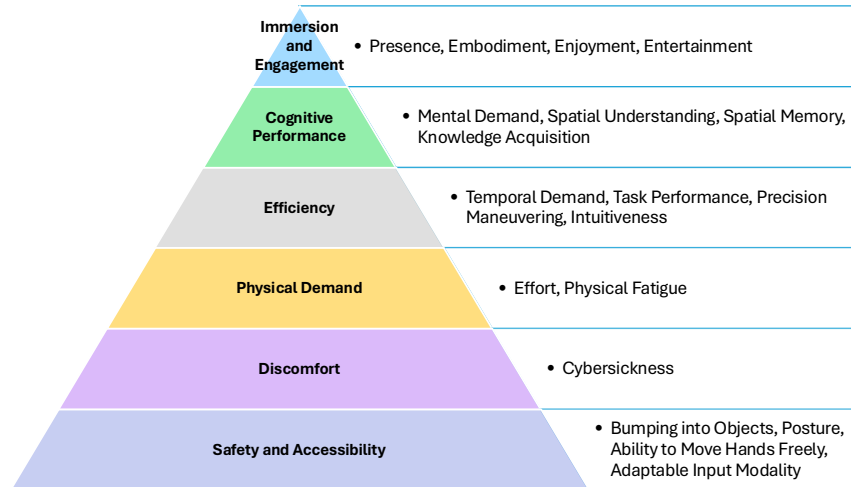


Figure 1: The preliminary framework consists of the following six categories that we believe encompass user experience when using virtual reality locomotion interfaces based on common evaluation metrics and user needs.

ABSTRACT

The evaluation of virtual reality locomotion interfaces has traditionally prioritized objective performance metrics like task completion time, while subjective factors like usability and accessibility remain underrepresented. As a result, a disconnect between design priorities and the practical needs of diverse users has emerged. To address this, my work develops a user-centric framework for assessing locomotion interfaces inspired by Maslow's Hierarchy of Needs. This work involves efforts to refine the preliminary model through expert engagement and empirical evaluations that map it to real user data. Ultimately, this work aims to bridge the gap between design priorities and practical application by enabling more inclusive, user experience-driven design for locomotion interfaces.

Index Terms: Locomotion, user experience, virtual reality.

1 INTRODUCTION

From a research perspective, the design of virtual reality (VR) locomotion interfaces has often prioritized developing universally optimal solutions across objective metrics such as task completion time and accuracy. In contrast, subjective evaluations of user experience and usability are under prioritized, evidenced by both ranking among the least explored metrics in locomotion research [8]. This limited usage of subjective evaluations might be due to the fact that they are simply difficult to measure [2]. Nevertheless, while objective metrics provide standardized benchmarks for comparison, they do not fully capture the holistic user experience. It is imperative to consider the user needs and issues of VR locomotion interfaces

due to their impact on user behavior [3]. For example, prior work has shown how distinct user groups respond differently to the same interface [6, 7]. Thus, not accounting for these needs can lead to suboptimal interfaces that are not equally accessible or effective for all users, revealing a misalignment in the practical needs of users and the design priorities of VR locomotion interfaces.

To address this gap, I introduce a framework that redefines the design priorities for locomotion interfaces based on user needs and issues. This framework takes inspiration from Maslow's Hierarchy of Needs [9] and its adaptations in computing, where it has been used to evaluate and explain end-user needs such as during long-term usage of head-mounted displays [4].

Maslow's Hierarchy of Needs is a psychology-based motivational theory that proposes the idea that humans are fueled by a desire to achieve goals and that reaching these goals is done by satisfying individual needs. It places foundational, short-term physiological needs at the base of the hierarchy and more psychological, long-term goals for personal growth towards the top. Through this organization of needs, the model explains why not meeting basic needs can make it difficult to pursue needs relating to personal growth and fulfillment. While hierarchical in nature, it is not strictly linear since multiple needs are oftentimes pursued simultaneously.

By grounding the framework in Maslow's Hierarchy of Needs, it centers user needs in the design priorities of locomotion interfaces by placing usability and accessibility at the forefront. As a result, it better supports long-term engagement and ensures that users of all backgrounds can use the interface without risk or exclusion. Furthermore, this framework also addresses the lack of user-centric classifications in existing locomotion taxonomies. Thus far, most taxonomies focus on creating classifications based on the attributes of an interface, the metaphor it invokes, or its degree of realism [10]. In contrast, the proposed framework considers the full spectrum of a technique's user-centric trade-offs, highlighting

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potential future directions for improved user experience, and classifying techniques based on their experiential strengths.

2 RESEARCH PROGRESS

In order to form the preliminary framework, I used an affinity diagramming approach to identify patterns and create themes from common evaluation metrics and user needs. This process began by compiling a list of all the metrics that have previously been used to evaluate locomotion interfaces, based on the recent survey on locomotion research trends by Martinez et al. [8]. I then expanded this initial list by incorporating the applicable sub-scales of each included survey. In addition, I integrated the user needs and issues identified by Hirobe et al. [5] as well as the accessibility needs identified by W3C [1]. The next step involved creating clusters of these items based on their similarities. I first considered the entire list of metrics and user needs, resulting in the following broad classifications: Physiological Needs, Usability and Performance, and User Experience. Then, I examined the items within each cluster to identify distinct sub-themes, resulting in the categories of the preliminary framework, seen in Figure 1.

Similar to Maslow's Hierarchy, the preliminary framework follows the idea of priority dependencies where foundational needs are a higher priority and must be met before the higher-level needs. Hence, categories with the strongest and most immediate influence on a user's ability to participate in locomotion tasks are of the highest priority and form the framework's foundation. The ordering of the categories mirrors how users actually experience locomotion: fundamental needs constrain participation, intermediate needs influence task success, and higher-level needs shape long-term satisfaction and engagement. The lower levels emphasize the fundamental barriers to participation such as Safety and Accessibility, and Discomfort. The middle levels focus on usability and performance through Physical Discomfort and Efficiency, needs that emerge once basic safety and comfort are established. The upper levels address the long-term, psychological needs by focusing on enhancing the satisfaction and overall quality of the experience through Cognitive Performance, and Immersion and Engagement.

3 FUTURE WORK

Building upon this preliminary framework, I have planned an interactive workshop session soliciting expert engagement for refinement that uses a thematic analysis approach. This workshop will follow an affinity diagramming format of collecting, grouping, and ordering ideas. Participants will be asked to list all of the metrics and user needs that are important to consider for any locomotion interface on sticky notes. Next, they will group these sticky notes by related ideas and summarize each cluster with a name. Finally, using these cluster names, participants will be asked to order the categories based on their perceived impact on user experience.

After refining the user needs hierarchy, I plan to conduct a systematic evaluation of existing locomotion interfaces applying the framework. Each interface will receive a "usability profile" that captures the full range of trade-offs in terms of user experience. These profiles will consist of scoring each individual category based on the extent the criterion are fulfilled. Then, a weighted total usability score will be calculated in which lower-level categories will have a higher weight, indicating their importance to the whole user experience.

The final stage of my thesis will focus on aligning the framework with real-user data through a series of empirical user study evaluations. The first study will focus on examining how the different categories of need interact with each other, looking at how unmet physiological needs, such as discomfort and fatigue, impact task and cognitive performance. In particular, it will be led by the following research question: *How do unmet physiological needs affect users' task performance and cognitive performance during*

VR locomotion? The second study will investigate whether task type (search, wayfinding, exploratory) influences the relative importance of each need category, led by the research question: *How does task type influence the perceived importance of each need category during VR locomotion?* Participants will complete search, wayfinding, and exploratory tasks using steering-based, walking-based, and selection-based interfaces. We will then evaluate how participants' perception of the user experience for each interface may vary depending on the task.

4 CONCLUSION

By following the research plan above, I aim to bridge the gap between locomotion interface design priorities and practical applications via a user-centric evaluation framework inspired by Maslow's Hierarchy of Needs. Ongoing efforts involve refining the current preliminary framework through expert engagement, applying that framework to assess established locomotion interfaces through a user-centric lens, and conducting a series of user studies to apply the framework to actual user datasets.

This framework places user needs at the center of locomotion interface design, thus supporting the development of more inclusive and accessible VR interfaces, and creating a user-centric classification system for locomotion techniques. By assessing locomotion interfaces through this framework, researchers and designers can more clearly identify usability trade-offs and uncover new directions for improving user experience. As this is ongoing work, there are still questions that require further thought. For example, I would like to discuss how to further hone the scoring structure of the usability profile as well as shaping the user study scope and design.

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